

Commerce XI Maths App

Exercise 3.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 30

Question 1 - Q1(i)

Write the conjugate of complex number $3 + i$

Options

- A. $3 - i$
- B. $-3 + i$
- C. $-3 - i$
- D. $3 + i$

Correct answer: $3 - i$

Explanation and steps

For $z = a + ib$, the conjugate is $a - ib$. Keep the real part unchanged and reverse the sign of the imaginary part.

Question 2 - Q1(ii)

Write the conjugate of complex number $3 - i$

Options

- A. $3 + i$
- B. $-3 - i$
- C. $-3 + i$
- D. $3 - i$

Correct answer: $3 + i$

Explanation and steps

For $z = a + ib$, the conjugate is $a - ib$. Keep the real part unchanged and reverse the sign of the imaginary part.

Question 3 - Q1(iii)

Write the conjugate of complex number $-\sqrt{5} - \sqrt{7}i$

Options

- A. $-\sqrt{5} + \sqrt{7}i$
- B. $\sqrt{5} - \sqrt{7}i$
- C. $\sqrt{5} + \sqrt{7}i$
- D. $-\sqrt{5} - \sqrt{7}i$

Correct answer: $-\sqrt{5} + \sqrt{7}i$

Explanation and steps

For $z = a + ib$, the conjugate is $a - ib$. Keep the real part unchanged and reverse the sign of the imaginary part.

Question 4 - Q1(iv)

Write the conjugate of complex number $-\sqrt{-5}$

Options

- A. $\sqrt{5} i$
- B. $-\sqrt{5} i$
- C. $\sqrt{5}$
- D. $-\sqrt{5}$

Correct answer: $\sqrt{5} i$

Explanation and steps

For $z = a + ib$, the conjugate is $a - ib$. Keep the real part unchanged and reverse the sign of the imaginary part.

Question 5 - Q1(v)

Write the conjugate of complex number $5i$

Options

- A. $-5i$
- B. $5i$
- C. 5
- D. -5

Correct answer: $-5i$

Explanation and steps

For $z = a + ib$, the conjugate is $a - ib$. Keep the real part unchanged and reverse the sign of the imaginary part.

Question 6 - Q1(vi)

Write the conjugate of complex number $\sqrt{5} - i$

Options

- A. $\sqrt{5} + i$
- B. $-\sqrt{5} - i$
- C. $-\sqrt{5} + i$
- D. $\sqrt{5} - i$

Correct answer: $\sqrt{5} + i$

Explanation and steps

For $z = a + ib$, the conjugate is $a - ib$. Keep the real part unchanged and reverse the sign of the imaginary part.

Question 7 - Q1(vii)

Write the conjugate of complex number $\sqrt{2} + \sqrt{3}i$

Options

- A. $\sqrt{2} - \sqrt{3}i$
- B. $-\sqrt{2} + \sqrt{3}i$
- C. $\sqrt{3} - \sqrt{2}i$
- D. $\sqrt{2} + \sqrt{3}i$

Correct answer: $\sqrt{2} - \sqrt{3}i$

Explanation and steps

For $z = a + ib$, the conjugate is $a - ib$. Keep the real part unchanged and reverse the sign of the imaginary part.

Question 8 - Q2(i)

Express the following in the form of $a + ib$,
 $a, b \in \mathbb{R}, i = \sqrt{-1}$. State the value of a and b .
 $(1 + 2i)(-2 + i)$

Options

- A. $-4 - 3i$; $a = -4, b = -3$
- B. $-4 + 3i$; $a = -4, b = 3$
- C. $-2 - 3i$; $a = -2, b = -3$
- D. $4 - 3i$; $a = 4, b = -3$

Correct answer: $-4 - 3i$; $a = -4, b = -3$

Explanation and steps

Replace every square root of a negative number by a multiple of i , use $i^2 = -1$, rationalise any denominator, and collect real and imaginary parts. This gives $-4 - 3i$; $a = -4, b = -3$.

Question 9 - Q2(ii)

Express the following in the form of $a + ib$,
 $a, b \in \mathbb{R}, i = \sqrt{-1}$. State the value of a and b .
 $(i(4 + 3i))/(1 - i)$

Options

- A. $-7/2 + (1/2)i$; $a = -7/2, b = 1/2$
- B. $7/2 - (1/2)i$; $a = 7/2, b = -1/2$
- C. $-1/2 + (7/2)i$; $a = -1/2, b = 7/2$
- D. $1/2 - (7/2)i$; $a = 1/2, b = -7/2$

Correct answer: $-7/2 + (1/2)i$; $a = -7/2, b = 1/2$

Explanation and steps

Replace every square root of a negative number by a multiple of i , use $i^2 = -1$, rationalise any denominator, and collect real and imaginary parts. This gives $-7/2 + (1/2)i$; $a = -7/2, b = 1/2$.

Question 10 - Q2(iii)

Express the following in the form of $a + ib$,
 $a, b \in \mathbb{R}, i = \sqrt{-1}$. State the value of a and b .
 $(2 + i)/((3 - i)(1 + 2i))$

Options

- A. $3/10 - (1/10)i$; $a = 3/10, b = -1/10$
- B. $3/10 + (1/10)i$; $a = 3/10, b = 1/10$
- C. $1/10 - (3/10)i$; $a = 1/10, b = -3/10$
- D. $3/5 - (1/5)i$; $a = 3/5, b = -1/5$

Correct answer: $3/10 - (1/10)i$; $a = 3/10, b = -1/10$

Explanation and steps

Replace every square root of a negative number by a multiple of i , use $i^2 = -1$, rationalise any denominator, and collect real and imaginary parts. This gives $3/10 - (1/10)i$; $a = 3/10, b = -1/10$.

Question 11 - Q2(iv)

Express the following in the form of $a + ib$,
 $a, b \in \mathbb{R}, i = \sqrt{-1}$. State the value of a and b .
 $(3 + 2i)/(2 - 5i) + (3 - 2i)/(2 + 5i)$

Options

- A. $-8/29 + 0i$; $a = -8/29, b = 0$
- B. $8/29 + 0i$; $a = 8/29, b = 0$
- C. $-4/29 + (19/29)i$; $a = -4/29, b = 19/29$
- D. $0 - (8/29)i$; $a = 0, b = -8/29$

Correct answer: $-8/29 + 0i$; $a = -8/29, b = 0$

Explanation and steps

Replace every square root of a negative number by a multiple of i , use $i^2 = -1$, rationalise any denominator, and collect real and imaginary parts. This gives $-8/29 + 0i$; $a = -8/29, b = 0$.

Question 12 - Q2(v)

Express the following in the form of $a + ib$,
 $a, b \in \mathbb{R}, i = \sqrt{-1}$. State the value of a and b .
 $(2 + \sqrt{-3})/(4 + \sqrt{-3})$

Options

- A. $11/19 + ((2\sqrt{3})/(19))i$; $a = 11/19, b = (2\sqrt{3})/(19)$
- B. $11/19 - ((2\sqrt{3})/(19))i$; $a = 11/19, b = -(2\sqrt{3})/(19)$
- C. $5/13 + ((2\sqrt{3})/(13))i$; $a = 5/13, b = (2\sqrt{3})/(13)$
- D. $2/19 + ((11\sqrt{3})/(19))i$; $a = 2/19, b = (11\sqrt{3})/(19)$

Correct answer: $11/19 + ((2\sqrt{3})/(19))i$; $a = 11/19, b = (2\sqrt{3})/(19)$

Explanation and steps

Replace every square root of a negative number by a multiple of i , use $i^2 = -1$, rationalise any denominator, and collect real and imaginary parts. This gives $11/19 + ((2\sqrt{3})/(19))i$; $a = 11/19, b = (2\sqrt{3})/(19)$.

Question 13 - Q2(vi)

Express the following in the form of $a + ib$,
 $a, b \in \mathbb{R}, i = \sqrt{-1}$. State the value of a and b .
 $(2 + 3i)(2 - 3i)$

Options

- A. $13 + 0i$; $a = 13, b = 0$
- B. $-5 + 0i$; $a = -5, b = 0$
- C. $4 - 9i$; $a = 4, b = -9$
- D. $13i$; $a = 0, b = 13$

Correct answer: $13 + 0i$; $a = 13, b = 0$

Explanation and steps

Replace every square root of a negative number by a multiple of i , use $i^2 = -1$, rationalise any denominator, and collect real and imaginary parts. This gives $13 + 0i$; $a = 13, b = 0$.

Question 14 - Q2(vii)

Express the following in the form of $a + ib$,
 $a, b \in \mathbb{R}, i = \sqrt{-1}$. State the value of a and b .
 $(4i^8 - 3i^9 + 3)/(3i^{11} - 4i^{10} - 2)$

Options

- A. $23/13 + (15/13)i$; $a = 23/13, b = 15/13$
- B. $23/13 - (15/13)i$; $a = 23/13, b = -15/13$
- C. $15/13 + (23/13)i$; $a = 15/13, b = 23/13$
- D. $-23/13 + (15/13)i$; $a = -23/13, b = 15/13$

Correct answer: $23/13 + (15/13)i$; $a = 23/13, b = 15/13$

Explanation and steps

Replace every square root of a negative number by a multiple of i , use $i^2 = -1$, rationalise any denominator, and collect real and imaginary parts. This gives $23/13 + (15/13)i$; $a = 23/13, b = 15/13$.

Question 15 - Q3

Show that $(-1 + \sqrt{3}i)^3$ is a real number. Its value is

Options

- A. 8
- B. -8
- C. $8i$
- D. $-8i$

Correct answer: 8

Explanation and steps

Write $-1 + \sqrt{3}i = 2(\cos 120^\circ + i \sin 120^\circ)$. Cubing gives $8(\cos 360^\circ + i \sin 360^\circ) = 8$, which is real.

Question 16 - Q4(i)

Evaluate i^{35}

Options

- A. $-i$
- B. i
- C. 1
- D. -1

Correct answer: $-i$

Explanation and steps

Powers of i repeat every four: $i^4 = 1$. Divide 35 by 4 and use the remainder to select 1, i , -1 , or $-i$.

Question 17 - Q4(ii)

Evaluate i^{888}

Options

- A. 1
- B. -1
- C. i
- D. $-i$

Correct answer: 1

Explanation and steps

Powers of i repeat every four: $i^4 = 1$. Divide 888 by 4 and use the remainder to select 1, i , -1 , or $-i$.

Question 18 - Q4(iii)

Evaluate i^{93}

Options

- A. i
- B. $-i$
- C. 1
- D. -1

Correct answer: i

Explanation and steps

Powers of i repeat every four: $i^4 = 1$. Divide 93 by 4 and use the remainder to select 1, i , -1 , or $-i$.

Question 19 - Q4(iv)

Evaluate i^{116}

Options

- A. 1
- B. -1
- C. i
- D. $-i$

Correct answer: 1

Explanation and steps

Powers of i repeat every four: $i^4 = 1$. Divide 116 by 4 and use the remainder to select 1, i , -1 , or $-i$.

Question 20 - Q4(v)

Evaluate i^{403}

Options

- A. $-i$
- B. i
- C. 1
- D. -1

Correct answer: $-i$

Explanation and steps

Powers of i repeat every four: $i^4 = 1$. Divide 403 by 4 and use the remainder to select 1, i , -1 , or $-i$.

Question 21 - Q4(vi)

Evaluate $(1)/(i^{58})$

Options

- A. -1
- B. 1
- C. i
- D. $-i$

Correct answer: -1

Explanation and steps

Powers of i repeat every four: $i^4 = 1$. Divide 58 by 4 and use the remainder to select 1, i , -1 , or $-i$.

Question 22 - Q4(vii)

Evaluate $i^{30} + i^{40} + i^{50} + i^{60}$

Options

- A. 0
- B. 1
- C. -1
- D. 2

Correct answer: 0

Explanation and steps

Powers of i repeat every four: $i^4 = 1$. Divide 30 by 4 and use the remainder to select 1, i , -1 , or $-i$.

Question 23 - Q5

Show that $1 + i^{10} + i^{20} + i^{30}$ is a real number. Enter its value

Correct answer: 0

Explanation and steps

Reduce each power using $i^4 = 1$: $i^{10} = -1$, $i^{20} = 1$ and $i^{30} = -1$. Hence $1 - 1 + 1 - 1 = 0$.

Question 24 - Q6(i)

Find the value of $i^{49} + i^{68} + i^{89} + i^{110}$

Options

- A. $2i$
- B. $-2i$
- C. 2
- D. 0

Correct answer: $2i$

Explanation and steps

Use the cycle $i, -1, -i, 1$. The four terms reduce to $i, 1, i$ and -1 , so their sum is $2i$.

Question 25 - Q6(ii)

Find the value of $i + i^2 + i^3 + i^4$

Correct answer: 0

Explanation and steps

$i + i^2 + i^3 + i^4 = i - 1 - i + 1 = 0$.

Question 26 - Q7

Find the value of $1 + i^2 + i^4 + i^6 + i^8 + \dots + i^{20}$

Correct answer: 1

Explanation and steps

The powers are $i^0, i^2, i^4, \dots, i^{20}$. They alternate 1 and -1 for eleven terms, leaving 1.

Question 27 - Q8(i)

Find the value of x and y , which satisfy the equation $(x + 2y) + (2x - 3y)i + 4i = 5$, where $(x, y \in \mathbb{R})$

Options

- A. $x = 1, y = 2$
- B. $x = 2, y = 1$
- C. $x = 3, y = 1$
- D. $x = 1, y = -2$

Correct answer: $x = 1, y = 2$

Explanation and steps

Equate real and imaginary parts: $x + 2y = 5$ and $2x - 3y + 4 = 0$. Solving gives $x = 1$ and $y = 2$.

Question 28 - Q8(ii)

Find the value of x and y , which satisfy the equation $(x + 1)/(1 + i) + (y - 1)/(1 - i) = i$, where $(x, y \in \mathbb{R})$

Options

- A. $x = -2, y = 2$
- B. $x = 2, y = -2$
- C. $x = 0, y = 0$
- D. $x = -1, y = 1$

Correct answer: $x = -2, y = 2$

Explanation and steps

Rationalise both denominators. Equating parts gives $x + y = 0$ and $y - x = 4$; therefore $x = -2$ and $y = 2$.

Question 29 - Q9(i)

Find the value of $x^3 - x^2 + x + 46$, if $x = 2 + 3i$

Correct answer: 7

Explanation and steps

Substitute $x = 2 + 3i$, expand using $i^2 = -1$, and combine like terms. All imaginary terms cancel and the value is 7.

Question 30 - Q9(ii)

Find the value of $2x^3 - 11x^2 + 44x + 27$, if $x = (25)/(3 - 4i)$

Correct answer: 2

Explanation and steps

First simplify $25/(3 - 4i) = 3 + 4i$. Substitute this value of x and use $i^2 = -1$; the expression simplifies to 2.